Aman Malik

Curriculum Vitae (Long Form)

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Profile

Areas of expertise
- **Energy System Modelling:** Scenario building and net-zero pathways development using models like GCAM and REMIND.
- **Carbon Markets & Policy:** Modelling-based research on India's Carbon Credit Trading Scheme (CCTS) and emission trading frameworks.
- **Transition Risk:** Climate-related financial risk assessments for the financial sector and institutional actors.
- **Sub-national Policy:** Developing state-level energy models and net-zero strategies (e.g., Gujarat and Uttarakhand).

Interests: Climate change mitigation policy; Indian energy policy; modelling energy and climate policy scenarios; employment implications of mitigation; climate-related financial risk; carbon dioxide removal.

Education

10/2017–11/2021 **Dr. rer. nat.**, *Technical University of Berlin and Potsdam Institute for Climate Impact Research*, Berlin and Potsdam, Germany

Title: Barriers to power sector decarbonisation in India

10/2014–05/2017 **Master of Science in Integrated Climate System Sciences**, *University of Hamburg*, Hamburg, Germany
Specialised in climate economics and carbon dioxide removal

Thesis Title: Kinetics of olivine dissolution in column experiments

Grade: 1.24 (*excellent*)

2009–2013 **Bachelor of Technology in Chemical Engineering**, *Guru Gobind Singh Indraprastha University*, New Delhi, India
First Division, *(72.4%)*

2009 **All India Senior Secondary School Examination**, *Springdales School*, New Delhi, India
93.2% (Indian Grade). Main courses: Chemistry, Mathematics, Physics, Computer Science and English

Additional Courses

08/2012 **An Introduction to the Science of Climate and Climate Change**, *University of Oxford, Department for Continuing Education (with climateprediction.net and the UK Met Office)*
Scored 100%

10/2012–01/2013 **Constructing and Analyzing High Resolution Climate Scenarios**, *University of Oxford, Department for Continuing Education*

02/2011 **GEG 4100 — Field Course in Glaciology of Antarctica**, *University of Ottawa*, Canada

2010 **ANTA 101 — Antarctica (online course)**, *Gateway Antarctica, University of Canterbury*, New Zealand

Work Experience

06/2023–02/2026 **Programme Lead**, *Council on Energy, Environment and Water*, New Delhi, India
Low Carbon Economy Team

10/2022–05/2023 **Consultant**, *Council on Energy, Environment and Water*, New Delhi, India

07/2022–12/2024 **Guest Scientist**, *Potsdam Institute for Climate Impact Research*, Potsdam, Germany

01/2022–06/2022 **Postdoctoral Researcher**, *Potsdam Institute for Climate Impact Research*, Potsdam, Germany

10/2017–12/2021 **Researcher**, *Potsdam Institute for Climate Impact Research*, Potsdam, Germany
Group of Energy Systems, Research Department III - Transformation Pathways

06/2017–08/2017 **Consultant**, *Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)*, New Delhi, India
NAMA implementation — Waste and Climate Change Adaptation in Rural Areas

10/2016–12/2016 **Student Assistant**, *Biogeochemistry Laboratory, Institute of Geology, University of Hamburg*, Hamburg, Germany
Data collection for a global geochemical database of ground and surface water

05/2016–09/2016 **Student Assistant**, *Climate Science and Policy Team, Climate Analytics*, Berlin, Germany
Development of SIAMESE — a model to emulate Integrated Assessment Models

02/2015–03/2016 **Student Assistant**, *Biogeochemistry Laboratory, Institute of Geology, University of Hamburg*, Hamburg, Germany
Working group of Prof. Dr. Jens Hartmann; used various instruments to measure ion concentration, surface area, alkalinity, pCO_2, and dissolved Si

04–06/2014 **Guest Writer**, *'Taleport' online Magazine (taleportmagazine.com), Happenstance Media*, New Delhi, India
Writing travel-based articles for the magazine

06–07/2012 **Intern**, *Gas Authority of India Limited*, Pata, Uttar Pradesh, India

2009–2011 **Team Mentor**, *Global Challenge Award*
Mentoring students from different teams in the online competition

Peer-reviewed Publications

2026 K. Dhandhania, **A. Malik**, R. Varma, V. Chaturvedi. "The Criticality of Country-Specific Scenarios to Inform Transition Risk Assessment for the Financial Sector: A case study for India". *Energy and Climate Change*. doi:10.1016/j.egycc.2026.100255

2025 P. Das and **A. Malik**. "Challenges to Deep-Decarbonization to Achieve Net Zero for India: A Review". *Strategic Analysis*.

2024 **A. Malik**, V. Chaturvedi, M. Sandhani, P. Das, C. Arora, N. Singh, R.Y. Cui, et al. "Implications of an emission trading scheme for India's net-zero strategy: a modelling-based assessment". *Environmental Research Letters*.

2022 T. Amann, J. Hartmann, R. Hellmann, E.T. Pedrosa, **A. Malik**. "Enhanced weathering potentials—the role of in situ CO_2 and grain size distribution". *Frontiers in Climate*.

2022 **A. Malik** and C. Bertram. "Solar energy as an early just transition opportunity for coal-bearing states in India". *Environmental Research Letters*.

2022 **A. Malik**. "Barriers to power sector decarbonisation in India". *PQDT Global* (Doctoral Dissertation).

2022 HL. van Soest... **A. Malik**, et al. "Global roll-out of comprehensive policy measures may aid in bridging emissions gap". *Nature Communications*.

2021 C. Bertram... **A. Malik**, et al. "Energy system developments and investments in the decisive decade for the Paris Agreement goals". *Environmental Research Letters*.

2021 L. Baumstark... **A. Malik**, et al. "REMIND2.1: Transformation and innovation dynamics of the energy–economic system within climate and sustainability limits". *Geoscientific Model Development*.

2021 **A. Malik**, C. Bertram, E. Kriegler, G. Luderer. "Climate policy accelerates structural changes in energy employment". *Energy Policy*.

2021 C. Bertram... **A. Malik**, et al. "COVID-19-induced low power demand and market forces starkly reduce CO_2 emissions". *Nature Climate Change*.

2020 **A. Malik**, C. Bertram, J. Després, J. Emmerling, S. Fujimori, A. Garg, et al. "Reducing stranded assets through early action in the Indian power sector". *Environmental Research Letters*.

2020 H. Wang, W. Chen, C. Bertram, **A. Malik**, E. Kriegler, G. Luderer, et al. "Early transformation of the Chinese power sector to avoid additional coal lock-in". *Environmental Research Letters*.

2019 F. Sferra, M. Krapp, N. Roming, M. Schaeffer, **A. Malik**, B. Hare, R. Brecha. "Towards optimal 1.5°C and 2°C emission pathways for individual countries: A Finland case study". *Energy Policy*.

Reports/Book-chapters

Aug 2026 **A. Malik**, V. Chaturvedi, R. Varma. "Transition Risk Scenarios for India's Financial Sector: Adapting Global Frameworks to National Realities and Financial Stability Risks". *CEEW Issue Brief*.

Nov 2025 CEEW, Ahmedabad University, and UNOPS. "Roadmap for a net zero power sector in Gujarat".

Apr 2025 CEEW, Ahmedabad University, and UNOPS. "Assessment of net zero scenarios for Gujarat".

Dec 2025 CEEW. "Assessment of net zero strategies for Uttarakhand" (Under review).

Dec 2024 P. Das, Z.A. Khan, ... A. Malik. "Supporting climate policy for Indian states: Learnings from long-term modelling". *CEEW Issue Brief*.

Working Papers & Preprints

2026 R. Varma, S. Pattnaik, **A. Malik**, V. Chaturvedi, J. Rajbanshi, D. Kumar, S.S. Srivastava. "Climate change impacts on the Indian energy system under net-zero pathways". *SSRN Working Paper* (Under Review). doi:10.2139/ssrn.6916596

2026 M. George, P. Das, V. Chaturvedi, **A. Malik**, et al. "Toward Next-Generation Climate Modeling in India: Lessons and Pathways for Action". *SSRN Working Paper*. doi:10.2139/ssrn.6197918

Policy Blogs

2026 **A. Malik**, S. Elango. "Why India must plan for AI data centres' energy and water footprint". *CEEW Blog*.

2023 **A. Malik**, N. Shukla. "How Energy System Modelling Tools Can Inform Design of India's Carbon Emission Trading Scheme". *CEEW Blog*.

Op-eds & Media

2024 **A. Malik**, V. Chaturvedi. "4 Factors to Influence Our Emissions Trading System". *New Indian Express*.

2024 **A. Malik**, P. Das. "Navigating India's Decarbonisation Challenges". *Moneycontrol*.

Positions, Awards, Honors and Recognitions

2024–2026 **Part of CEEW's team**, *Expert Committee on Reserve Bank's Climate-related Information System*

2023 **IOP Trusted Reviewer**, *In recognition of exceptionally high level of peer review consistency*, IOP Publishing

11/2023 **Full Scholarship**, *Integrated Assessment Modelling Consortium Annual Conference*, Venice, Italy
Fully-funded scholarship from EU DG CLIMA to present paper on carbon markets

12/2019–03/2020 **Guest exchange scholarship**, *COMMIT project*, PBL, Netherlands
Fully funded 3-month long exchange at TERI, New Delhi, India

06/2019–07/2019 **Participant**, *CD-LINKS Summer School*, Venice, Italy
Fully-funded attendance at a week-long summer school on Integrated Assessment Models

2013 **2nd position**, *'Chem-E-Car' competition, Cognizance (annual tech fest)*, IIT Roorkee, India
Organised by the Chemical Engineering Department; cash prize Rs. 5000

2013 **2nd position**, *'Chain Reaction Contraption' competition*, IIT Roorkee, India
Cash prize Rs. 4000

2012 **3rd position**, *'Chem-E-Car' competition, Cognizance (annual tech fest)*, IIT Roorkee, India
Organised by the Chemical Engineering Department; cash prize Rs. 3000

04/2014 **Consolation Prize**, *Jawahar Institute of Mountaineering*, Pahalgam, India
For being one of the best students in the mountaineering course

04–09/2016 **Merit scholarship worth 3250€**, *University of Hamburg, Germany*

06–08/2013 **Summer Fellowship**, *Jawaharlal Nehru Center for Advanced Scientific Research*, Bangalore, India
Supervisor: Prof. Dr. Roddam Narasimha

10/2012–01/2013 **Bursary of £1250**, *University of Oxford, Department of Continuing Education*
Scholarship for online course titled 'Constructing and Analyzing High Resolution Climate Scenarios' (8 Oct 2012 – 4 Jan 2013)

02/2011 **Expedition Grant worth US$ 8500**, *for the 'Students on Ice' Antarctic University Expedition (uantarctic.org)*
14–27 February 2011. Grants provided by Guru Gobind Singh Indraprastha University, National Centre for Antarctic and Ocean Research, and the Council for Scientific and Industrial Research

2011 **3rd position**, *Brand's Quiz, 'Infoexpression' (annual fest, University School of Information Technology)*, GGSIP University, India

2011 **3rd position**, *General Quiz, 'Infoexpression' festival, University School of Information Technology*, GGSIP University, India

2010 **2nd position**, *Science Quiz, 'Pulse 2010' Literary Fest*, All India Institute of Medical Sciences, India
Cash prize Rs. 1000

2009 **Certificate of Merit**, *Central Board of Secondary Education*
For outstanding academic performance and for being among the 0.1% of all successful candidates in Physics in the All India Secondary School Examination

2009 **Full Scholarship**, *GIV Engineering Program*, University of Vermont
Fully funded scholarship to attend an engineering summer school

2009 **Best in Class**, *GIV Engineering Program*, University of Vermont, USA
Award for being the best design team in the theme Biomass

2009 **Summer Fellowship**, *Mathematical Sciences Foundation*, New Delhi, India

2008–09 **2nd position**, *Global Business Plan category, Global Challenge Award*, USA
Topic: Carbon dioxide capture through Zeolitic Imidazolate Frameworks

2007–08 **3rd position**, *Global Business Plan category, Global Challenge Award*
Cash prize $900. Topic: Generating electricity by working up a sweat

2007–08 **3rd position**, *Best STEM Explorations category, Global Challenge Award*

Extracurricular

2018/2019 **Section E Officer**, *Leibniz Association PhD Network*

2018/2019 **PhD Representative**, *Potsdam Institute for Climate Impact Research*

2015/2016 **Delegate**, *University of Hamburg*, UNFCCC COP 21, Paris and COP 22, Marrakesh

04/2015–09/2016 **Event Organizer**, *PIASTA — Intercultural Living and Learning, University of Hamburg*, Hamburg, Germany

05/2015–06/2016 **Member, Student Council**, *School of Integrated Climate System Science, University of Hamburg*, Hamburg, Germany

05/2015–06/2016 **Student Member, Examination Committee**, *School of Integrated Climate System Science, University of Hamburg*, Hamburg, Germany

2014 **Presenter**, *Photo Fest 2014, Anugoonj — Intercultural Festival of Indraprastha University*, New Delhi, India

04/2014 **Basic Course in Mountaineering**, *Jawahar Institute of Mountaineering*, Pahalgam, Jammu & Kashmir, India

01–02/2013 **Basic Course in Skiing**, *Atal Bihari Vajpayee Institute of Mountaineering and Allied Sports*, Manali, Himachal Pradesh, India
27 January – 9 February 2013

2012 **Head Coordinator, Literary Events**, *'Anugoonj' — Annual Cultural Festival*, Indraprastha University, New Delhi, India

03/2012 **Contributor**, *'Geography & You' magazine*, India
Contributed photographs to the special Polar Region issue

2010 **Volunteer**, *King Cobra Telemetry Project*, Agumbe Rainforest Research Station, Karnataka, India

2009–2010 **Project Member**, *Corporate Section, Indian Youth Climate Network*, New Delhi, India
Organising events for companies as part of their Corporate Social Responsibility policy

05/2009 **Author**, *'No Wake Up Call but a Screaming Siren'*, Global Challenge Award Newsletter

Book co-author, *Pal, R., Solanki, S., &* ***Malik, A.***, (2013). Wings of Dwarka — a field guide to bird watching, New Delhi: Guru Gobind Singh Indraprastha University

In the Media

13/05/2011 **Newspaper feature**, *'Aman Malik gets an opportunity to go to the Polar Regions', Amar Ujala*, Youth Section, p. 7

16/05/2011 **Newspaper feature**, *'IP boy visits Antarctica for research'*, Tribune News Service

17/05/2011 **Newspaper feature**, *'IP University Student tries to understand the mysteries of Antarctica' (translation)*, Hindustan, p. 4

18/05/2011 **Newspaper feature**, *'IP University student carries out research in Antarctica', HT Horizons*, Hindustan Times, p. 6

24/05/2011 **Newspaper feature**, *'Indian student in Antarctica', Education Mail*, Mail Today, p. 3

26/05/2011 **Newspaper feature**, *'On top of the world'*, The Pioneer

16/06/2011 **Newspaper feature**, *'Conquering Ice', Education Times*, Times of India

30/03/2011 **Radio interview**, *All India Radio — Young Scientist Program ('Yuva Bharti', Youth Channel)*

10/2009 **Interview**, *cogito.org, John Hopkins Center for Talented Youth*, John Hopkins University

Review work

Reviewer for the following journals:

Nature

Environmental Research Letters: Climate

Climate Policy

Energy Policy

Climate Economics

Energy Research and Social Science

Environmental Research Letters

Computer Skills

proficient MS Office, R

good Python, QGIS

basic GAMS

Languages

Hindi **native**

English **near-native fluency**

German **very good, level C1**

Talks, Lecture, Panel discussions

09/2026 **Guest lecture**, *Energy Transition and Decarbonisation*, IOCL Senior Management Programme with IIM Calcutta

04/2026 **Guest lecture**, *Economics and Geopolitics of Energy*, WWF-India and OP Jindal University, Students of LL.M. ELECC 2025

02/2026 **Invited talk**, *The Emerging Carbon Markets Landscape in India*, Centre for Nature-based Climate Solutions, National University of Singapore

12/2025 **Panelist**, *Session on carbon removal in Article 6*, CO2 Forward Carbon Removal Summit, Carbon Removal Alliance of India
Delhi

10/2025 **Guest lecture**, *Economics of Climate Change*, WWF-India and OP Jindal University, Students of LL.M. ELECC 2025-2026

10/2025 **Moderator**, *Panel discussion on Decarbonisation of Economy – Leveraging Carbon Credits*, Viksit Gujarat Regional Summit, Ganpat University
Mehsana

08/2025 **Guest lecture**, *Energy Transition and Decarbonisation*, IOCL Senior Management Programme with IIM Calcutta

07/2025 **Speaker**, *India Emissions and Net Zero*, Indian Energy Forum, New Delhi

05/2025 **Guest lecture**, *Decision maker under uncertainty: Insights from modelling*, National Thermal Power Corporation

03/2025 **Panelist**, *Panelist at Sustainable Futures Collaborative's Tufts Climate Panel Discussion*, New Delhi

01/2025 **Guest lecture**, *Introduction to Modelling*, IIT Roorkee

09/2024 **Speaker**, *Conference on Joint Response to Climate Crisis*, Seoul, South Korea
Sookmyung Women's University

09/2024 **Speaker**, *Lecture on Climate Change at the Rajya Sabha (Upper House) for Parliamentary Interns*, New Delhi, India
To interns of Members of Parliament

09/2024 **Speaker**, *IIT Delhi workshop on Energy and Well-being – Indian Case Study*, New Delhi, India
IIT Delhi

09/2024 **Speaker**, *Surya Foundation Think Tank Strengthening Some Key Areas for Viksit Bharat 2047*, New Delhi

08/2024 **Speaker and Panelist**, *Carbon Markets in India*, New Delhi, India
Centre for Social and Economic Progress

08/2024 **Speaker**, *State Dialogue on Accelerating Energy Efficiency in Gujarat*, Vadodara, India
[Watch Video]

02/2024 **Guest speaker**, *Webinar on Role of ETS in India's Net-Zero targets*, Delhi, India
Organised by the Carbon Markets Association of India

02/2024 **Expert speaker**, *Accelerating the Just Transition of MSMEs: Capacity Development of Policymakers Program*, Lucknow, India
Organised by School of Public Policy and UNDP

01/2024 **Panelist**, *Impact of Net-Zero Transition Policies on Clean Jobs, Skilling Needs and Interstate Migration*, New Delhi, India
Event organised by ICRIER — [Watch Video]

01/2024 **Speaker**, *Implications of India's ETS on its net-zero target*, New Delhi, India
Niti Aayog

11/2023 **Oral presentation**, *Integrated Assessment Modelling Consortium*, Venice, Italy
Role of ETS in India's Net-zero targets: A modelling-based assessment

03/2022 **Defense presentation**, *Barriers to power sector decarbonisation in India*, TU Berlin, Germany

12/2020 **Oral presentation**, *Integrated Assessment Modelling Consortium*, Online
Assumptions on employment factor approach in energy scenarios

Aman Malik and Christoph Bertram

09/2020 **Expert Panel**, *Climate Action and Renewable Energy in context of a Green Recovery*, Africa–EU Climate Partnership, Online

07/2019 **Presentation**, *Preventing stranded assets and empowering renewables*, CD-LINKS Summer School, Venice
Italy

11/2018 **Poster presentation**, *Integrated Assessment Modelling Consortium*, Seville, Spain
'Not (yet) another coal story: Power-sector lock-ins in India - perspective from national and global models'
Aman Malik and Christoph Bertram